Radial Basis Function (RBF) Kernel

$$e^{-\frac{1}{2}(a-b)^2} = e^{-\frac{1}{2}(a^2+b^2-2ab)} = e^{-\frac{a^2}{2}} e^{-\frac{b^2}{2}} e^{ab} =$$

$$e^{-\frac{a^2}{2}}\left(1,\ \frac{a}{\sqrt{1!}},\ \frac{a^2}{\sqrt{2!}},\ \frac{a^3}{\sqrt{3!}},\ \frac{a^4}{\sqrt{4!}},\ \ldots.,\ \ldots.,\ \infty\right)$$

$$\phi(a)^T$$

$$\phi(b) \qquad e^{-\frac{b^2}{2}}$$

$$\begin{vmatrix} 1 \\ \frac{b}{\sqrt{1!}} \\ \frac{b^2}{\sqrt{2!}} \\ \frac{b^3}{\sqrt{3!}} \\ \frac{b^4}{\sqrt{4!}} \\ \ldots. \\ \ldots. \ \infty \end{vmatrix}$$

RBF Kernel

$$e^{-\gamma(a-b)^2}$$